Predicting First-Year Statistics Grades

From Attendance, Honesty, and Quiz Results

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Contents

Research Hypotheses

H0 — Behavioural predictors vs. attitude

Attendance, homework completion/timeliness, and academic honesty (`DISS`) predict midterm scores more strongly than attitude toward mathematics (`atitMATH`).

H1 — Peer-proximity effects

Seating near higher-achieving peers ($\mathtt{PROXI}_{75}$) raises scores; seating near lower-achieving peers ($\mathtt{PROXI}_{25}$) may depress them — both relative to the median student.

H2 — Homework orderliness & quiz performance

Lower attendance, late/digital submission, low quiz scores, and greater perceived homework difficulty are jointly associated with lower midterm results.

Model Specifications (H0)

First midterm Y_1:

$$Y_1 = \alpha + \beta_1 \, \text{GENDER} + \beta_2 \, Y_1\text{GROUP} + \beta_3 \, \overline{\text{ATND}}_1 + \beta_4 \, \text{numHW}_1 + \beta_5 \, \text{paperHW}_1 + \beta_6 \, \text{lateHW}_1 + \beta_7 \, \text{DISS}_1 + \beta_8 \, \text{atitMATH}$$

Second midterm Y_2 (reduced — lower HW variance):

$$Y_2 = \alpha + \beta_1 \, \text{GENDER} + \beta_2 \, Y_2\text{GROUP} + \beta_3 \, \overline{\text{ATND}}_2 + \beta_4 \, \text{numHW}_2 + \beta_5 \, \text{DISS}_2 + \beta_6 \, \text{atitMATH}$$

Autoregressive extensions add `firstQIZZ` (for Y_1) or $\widehat{Y_1}$ (for Y_2); 2SLS instruments the prior score for endogeneity.

Model Specifications (H1)

Baseline proximity models (both midterms):

$$\begin{aligned} Y_{1/2} = \alpha &+ \beta_1\,\text{GENDER} + \beta_2\,\text{GROUP} \\ &+ \beta_3\,\text{PROXI25} + \beta_4\,\text{PROXI75} \end{aligned}$$

Dishonesty + interaction sub-hypothesis (disentangling cheating from learning):

$$\begin{aligned} Y_{1/2} = \alpha &+ \beta_1\,\text{GENDER} + \beta_2\,\text{GROUP} + \beta_3\,\text{DISS} \\ &+ \beta_4\,\text{PROXI25} + \beta_5\,\text{PROXI75} + \beta_6\,\text{INT}(\text{DISS},\,\text{PROXI75}) \end{aligned}$$

Note: `INT(DISS, PROXI25)` omitted — no plausible cheating mechanism from lower-achieving neighbours.

Sample

- **49 regular students** enrolled in *Statistics for Economists* (4EF113421), all five majors, Faculty of Economics — Goce Delchev University, Shtip.
- Only **30 of 49** students completed the first knowledge-assessment quiz (`firstQIZZ`); a large share of the cohort were exam-only registrants.
- Data collected: paper and evidence-list **attendance**, per-session **quiz scores**, paper-list **seating index**, and multiple **dishonesty indicators**.
- Hybrid time-cross-sectional design with **two non-equivalent midterm periods** (Y_1 and Y_2); pooling additional cohorts was infeasible.
- *Key limitation:* Small N constrains statistical power and external validity; results are suggestive rather than definitive.

Experimental Design

- **Non-disclosure:** Procedures were not revealed to participants so as to preserve natural behaviour.
- **Quizzes** administered via Google Forms (budget constraint precluded timed platforms such as Kahoot). Each quiz was framed as optional self-assessment not affecting grades.
- **Triple attendance check:** paper list, evidence list, and quiz-participant list — consistency across three independent records made coordinated manipulation substantially harder.
- **Attention/honesty traps:** Default quiz responses set to "Yes" (submitted homework) and "3" (moderate difficulty); unchanged defaults flagged inattention or dishonesty.
- **Privacy:** Final dataset fully anonymised via `anonINDX`; no names retained.

Feature Engineering: Identifiers & Background

- **anonINDX** *[cont]* — ascending anonymisation index; also serves as a control for random enrolment effects.
- **GENDER** *[log]* — inferred from first name: $0 =$ male, $1 =$ female; not directly surveyed.
- **HSCH** *[categ, 1–3]* — secondary school type (self-reported): $1 =$ general, $2 =$ economics/law specialised, $3 =$ other specialised.
- **MJR** *[categ, 1–6]* — enrolled major: Finance, Business Economics, Marketing & Management, Accounting & Audit, International Economics & Business, Other.
- **atitMATH** *[categ, 1–6]* — self-rated attitude to mathematics ($1 =$ dislike very much $\rightarrow 6 =$ favourite subject).

Feature Engineering: Attendance & Quizzes

- **averQIZZ**$_{1/2}$ *[cont, 0–1]* — average quiz score across sessions in period 1 / period 2.
- **diffQIZZ**$_{1/2}$ *[cont, 0–1]* — difference between first-quiz score and period average (captures learning trajectory).
- **averATND**$_{1/2}$ *[cont]* — composite attendance rate, averaged from:
  - `numP` — paper-list attendances
  - `numE` — evidence-list attendances
  - `numQ` — quizzes taken
- `_H` suffix — binary indicator: 1 if student's value exceeds sample mean; `c_` prefix — mean-centred version (for multicollinearity control).

Feature Engineering: Dishonesty Index (DISS)

- Cumulative index; $\texttt{DISS}_2 = \texttt{DISS}_1 +$ new incidents in period 2.

Component weights

Component	Variable	Points
Falsified evidence-list signature	`honLST`	+3
Suspicious midterm behaviour	`Y1/Y2CHEAT`	+2
Implausible homework claim	`honHW`	+1

- `CHEAT` denotes behaviours *indicative of* cheating but not conclusive enough for the university's standard sanction (automatic fail + one-year retake ban).

Feature Engineering: Homework & Seating Proximity

Homework variables:

- **numHW**$_{1/2}$ — number of homeworks submitted
- **paperHW**$_{1/2}$ — proportion submitted by hand (vs. digital)
- **lateHW**$_{1/2}$ — proportion submitted after deadline
- **hardHW**$_{1/2}$ — average perceived difficulty (rescaled 0–1 from 1–6 scale)
- **additHW**$_{1/2}$ — additional (course-project) homework submitted

Seating proximity:

- **PROXI25**$_{1/2}$ — fraction of sessions within ± 2 index positions of a peer *below the 25th percentile*
- **PROXI75**$_{1/2}$ — fraction of sessions within ± 2 index positions of a peer *above the 75th percentile*

$\Rightarrow$ Index positions proxy for physical bench proximity (rows of two benches, four seats per connected unit); averaged across sessions to reduce outlier influence.

Baseline Model: Predicting `firstQIZZ`

$$\text{firstQIZZ} = \alpha + \beta_1\,\text{GENDER} + \beta_2\,\text{anonINDX} + \beta_3\,\text{HSCH} + \beta_4\,\text{MJR} + \beta_5\,\text{atitMATH}$$

OLS results ($N = 25$ of 49)

Variable	Coeff.	Std. Err.	p-value
Constant	0.259	0.122	0.046
anonINDX	0.003	0.002	0.255
GENDER	0.098	0.071	0.181
HSCH	0.005	0.042	0.902
MJR	−0.043	0.032	0.192
atitMATH	−0.006	0.023	0.805

$R^2 = 0.203$, Adj. $R^2 = -0.007$, F-stat $p = 0.463$

Model is not statistically significant. Likely causes: small N, omitted prior

H0: Dishonesty vs. Attitude Toward Math

Key finding

Each one-point increase in `DISS` is associated with an average decline of **7.6–11.2 percentage points** in midterm scores, across all OLS, autoregressive, and 2SLS specifications.

- `DISS` coefficients: -0.077 to -0.112 (all $p < 0.05$ in OLS/AR); effect persists in Y_1 and Y_2.
- `atitMATH` is **consistently insignificant** across all nine specifications ($p > 0.26$); coefficients hover near zero.
- `averATND` occasionally significant but *opposite sign* to expectation — likely reflects reverse causality or multicollinearity.
- `numHW` shows occasional significance for Y_2 but is unstable.
- **Conclusion:** H0 supported — behavioural/honesty measures outperform attitudinal ones.

Regression Results for H0 (summary)

Selected coefficients for DISS and atitMATH

Model	DISS coeff.	p	atitMATH coeff.	p
1H0 (OLS, Y_1)	-0.091	0.004^{***}	0.020	0.270
1H0_ar (Y_1 + AR)	-0.095	0.004^{***}	0.016	0.513
1H0_red (OLS, Y_1)	-0.077	0.010^{***}	0.020	0.262
2H0 (OLS, Y_2)	-0.112	0.013^{**}	0.024	0.386
2H0_ar (Y_2 + AR)	-0.083	0.059^{*}	0.008	0.776

H1: Seating Proximity and Peer Effects

Key result — PROXI75 (near higher achievers)

Consistently **positive and significant** across OLS specifications for both midterms. Effect remains after controlling for DISS and persists in Y_2 (when collusion was effectively prevented).

- PROXI75 Y_1 OLS: $\hat{\beta} = +0.142$, $p = 0.035^{**}$
- PROXI75 Y_2 OLS: $\hat{\beta} = +0.488$, $p = 0.006^{***}$
- Interaction term INT(DISS, PROXI75) is significant for Y_2 (coeff. -0.132, $p = 0.034^{**}$), suggesting some collusion component — but the main effect of PROXI75 survives.
- **PROXI25**: No consistent or significant effect in any specification — proximity to lower-achieving peers does not detectably harm performance.
- **Interpretation:** Knowledge transfer, social learning, or increased effort — not mere copying.

H2: Homework Orderliness & Performance

Result: H2 not supported

Coefficients for attendance (`averATND_H`), homework timeliness (`lateHW`), format (`paperHW`), quiz performance (`averQIZZ_H`), and perceived difficulty (`hardHW_H`) are **statistically insignificant throughout**.

- Despite some models reaching overall significance (F-stat $p < 0.05$), no individual H2 predictor is reliably significant.
- `averATND_H` appears with *negative* sign — contrary to expectations; likely an artefact of small N and measurement noise.
- **Null results do not close the question:** homework was not rubric-scored; objective quality measures were unavailable.
- Future work: standardised grading rubrics, automated participation logs, student question-rates as engagement proxies, nonlinear model exploration.

Discussion: Main Findings

- **H0 — Supported:** `DISS` is a strong, robust negative predictor (-7.6 to -11.2 pp per point) across all specifications; `atitMATH` is uniformly insignificant.
- **H1 — Supported for `PROXI75`:** Sitting near higher-achievers raises scores; the effect survives dishonesty controls and the harder $Y_2 exam$, suggesting genuine peer learning.
- **H1 — Not supported for `PROXI25`:** No evidence that lower-achieving neighbours hurt performance.
- **H2 — Not supported:** The hypothesised linear pathway (attendance $\rightarrow$ homework habits $\rightarrow$ quiz $\rightarrow$ exam) is undetectable in the current small sample.
- **Policy implication:** Strategic seating arrangements could be a *low-cost* classroom intervention to boost early-course achievement.
- **Caution:** $N = 49$, single cohort, single instructor — results are suggestive; causality not definitively established.

Limitations

- **Sample size:** Only 49 regular students; fewer complete cases in some specifications. Small N limits statistical power and external validity.
- **Measurement quality:** Homework was not rubric-scored for sufficiency/correctness; dishonesty proxies rely on observable behaviour, not conclusive evidence.
- **Observer effects:** One individual recorded pre-test behaviour, observed in-test behaviour, *and* later participated in grading after an operational change — potential bias despite mitigation steps.
- **Covert behaviour:** Undetected cheating and awareness-induced behaviour changes remain possible; unobtrusive recording would be preferable but raises ethical and logistical concerns.
- **Omitted variables:** Prior mathematics achievement (prerequisite grade) was unavailable; its exclusion may bias estimated relationships.

Future Research Directions

- **Scale up:** Replicate across larger, more heterogeneous samples, multiple cohorts, courses, and institutions.
- **Objective measures:** Rubric-based homework scoring; collect standardised prior-ability indicators (e.g., prerequisite course grade).
- **Tighter protocols:** Separate data collection from grading; pre-register procedures; use independent proctoring or automated monitoring where ethically permissible.
- **Dishonesty robustness:** Exclude or instrument for flagged observations; use validated behavioural measures of cheating.
- **Advanced modelling:** Test non-linearities and interaction effects (e.g., proximity $\times$ ability); employ panel or mixed-effects models with longitudinal data to better isolate causal mechanisms.
- **Additional proxies:** Student question-rates, lecture attention metrics, participation counts during exercises.

Questions and Discussion

I appreciate your attention.

Q & A?